My research interest includes World Models, Reinforcement Learning, Diffusion Models, and Knowledge Distillation.

In my project, "World Model-Guided Multi-Agent Reinforcement Learning," I addressed the sample inefficiency and high trial-and-error costs associated with deploying RL in physical environments. Considering that edge devices in the same environment follow the same physical laws, I leveraged the historical interaction trajectories of all agents to train a centralized World Model (WM). This WM generates multi-step predictions from current states and actions, guiding policy exploration and accelerating MARL convergence. Furthermore, to tackle the growing exploration space inherent to multi-agent settings, I introduced fine-grained constraints on the action space during the training phase. Extensive evaluations demonstrate that this framework not only yields superior policies but also accelerates convergence by approximately 36% compared to baseline algorithms. This project has yielded a first-author paper published at *VTC 2025 Fall*, with an extended manuscript currently under review at the *IEEE Internet of Things Journal (IoTJ)*.

Parallel to my work in environment modeling and RL, my research in Diffusion Model-Assisted Data-Free Knowledge Distillation (DFKD) explores how to efficiently deploy complex deep learning models on resource-constrained edge devices. While KD is a powerful model compression paradigm, its traditional reliance on original training data poses privacy challenges. To address this, most existing DFKD studies focus on improving the diversity of synthesized samples, often overlooking whether the student model acquires core semantic knowledge in these synthesized samples. I proposed a novel diversity-enhancement framework based on attention constraints. By extracting the teacher model's regions of interest from synthetic images generated via GANs, I applied CutMix and MixUp to the background areas, constraining the student model's attention to align with the teacher's focus. Due to the lack of semantic information in the images synthesized by GANs, I employed GLIDE to augment the synthetic dataset. These methods ensure effective knowledge transfer without original data, achieving state-of-the-art (SOTA) performance on CIFAR-10/100. The research findings from this project are planned for submission to *AAAI 2027*.

Building upon my master's research experience in both efficient decision-making and model compression, I am eager to pursue doctoral research centered on the efficient inference of world models and self-evolving embodied agents. First, considering that robots are constrained by limited computational resources yet require real-time responses to complex environments, I aim to explore how to optimize world models for lightweight and efficient inference. Second, as edge agents continuously accumulate experiential data through real-world interactions, I hope to investigate how to leverage this streaming data to drive self-evolution and continuous adaptation. I believe this is a critical path to enhancing the robustness and generalization capabilities of agents in dynamic environments.